

- The most important feature: Persistent Storage (PS) is now supported to increase battery life, and battery power is also reserved to maintain data in the RAM to avoid any loss of data due to power failure.

5.1.3 Linux On Mobile

The open source community's most talked-about offer to the computing world is the Linux OS, be it desktops or microcontrollers. The idea behind open source is to give anyone the freedom to download the source code, learn, debug and develop new code for themselves and the community. This greatly helps the debugging process of an OS, thus providing a robust, error-free OS. Developing OSes are tagged as 'unstable versions', which means that the debugging stage is still incomplete. The final release is tagged as the 'stable version'. The same thing goes for desktops as well as microdevices such as mobile phones.

Mobile manufacturers are making efforts to develop better Linux versions for mobile, or rather telecommunication, devices. As Linux is open source, they do not have to purchase a license. This will also result in a price competition, as the cost of production of Linux-based mobile phones will be lower, so more features can be provided to the buyer.

Developments to this effect happened independently, and the movement did not gather much momentum. As a result, there were few mobile phones powered by Linux. Open Source Development Labs (OSDL), an industry consortium, has taken the initiative to spur the development of Mobile Linux and related applications, and host open source projects for the same.

The first highly optimised Linux OS for smartphones, called Mobilinux, was developed by MontaVista Software, a company in California that develops system software and embedded Linux for consumer and automotive electronics. Based on open source and open standard (like Symbian), Mobilinux was devised to provide scalability and maximise the battery life of single-chip mobile phones. The Kernel used is 2.6, and has a boot time of less than one